

Challenge (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 16
- (C) 32
- (D) 4
- (E) 6

Q2. Convert $\sqrt[3]{8}$ to rational exponent form

- (A) $2^{1/2}$
- (B) $2^{1/3}$
- (C) 2^3
- (D) $2^{2/3}$
- (E) $8^{1/3}$

Q3. In a gaming tournament, a player's score is $\sqrt{256}$. What is the score?

- (A) 12
- (B) 16
- (C) 10
- (D) 14
- (E) 8

Q4. Simplify $\sqrt{100}$

- (A) 5
- (B) 10
- (C) 20
- (D) 50
- (E) 25

Q5. A gaming computer uses $\sqrt[3]{27}$ GB of RAM. How many GB does it use?

- (A) 2
- (B) 3
- (C) 6
- (D) 9
- (E) 12

Q6. Convert $8^{1/3}$ to radical form

- (A) $\sqrt[3]{8}$
- (B) $\sqrt{8}$
- (C) $\sqrt[4]{8}$
- (D) $\sqrt[2]{8}$
- (E) $\sqrt[5]{8}$

Q7. A mobile game uses $\sqrt{121}$ MB of data. How many MB does it use?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q8. Simplify $\sqrt{49}$

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 5

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{256} + \sqrt[3]{64}$. Show your work and provide the final answer.

Q10. A gaming company uses the formula $\sqrt[3]{x}$ to calculate the score of a player, where x is the number of points earned. If a player earns 216 points, what is their score? Show your work and provide the final answer.

Chef's Tips

- Tip 1: Emphasize the importance of simplifying radicals to their simplest form
- Tip 2: Encourage students to use rational exponent form to simplify complex radicals
- Tip 3: Use real-world examples, such as gaming and technology, to illustrate the application of radicals